

SCADA

Designed By

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Problem Statement

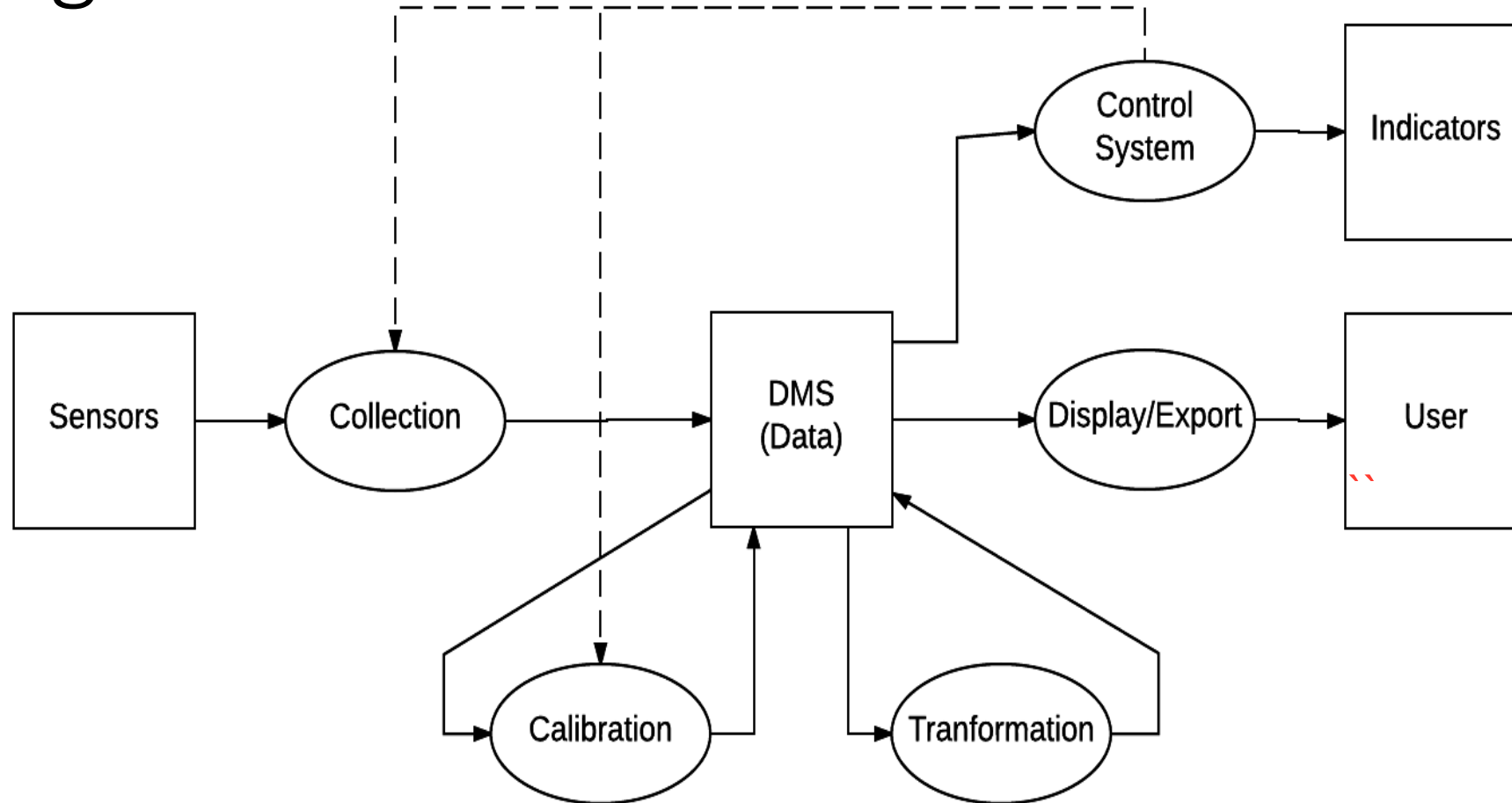
This project deals with designing a Supervisory Control and Data Acquisition (SCADA) system that has high configurability.

- Record Sensor Data
- Configurable sensor hierarchy
- Configurable data acquisition states
- Configurable calibration of data
- Exporting data

Supported Features

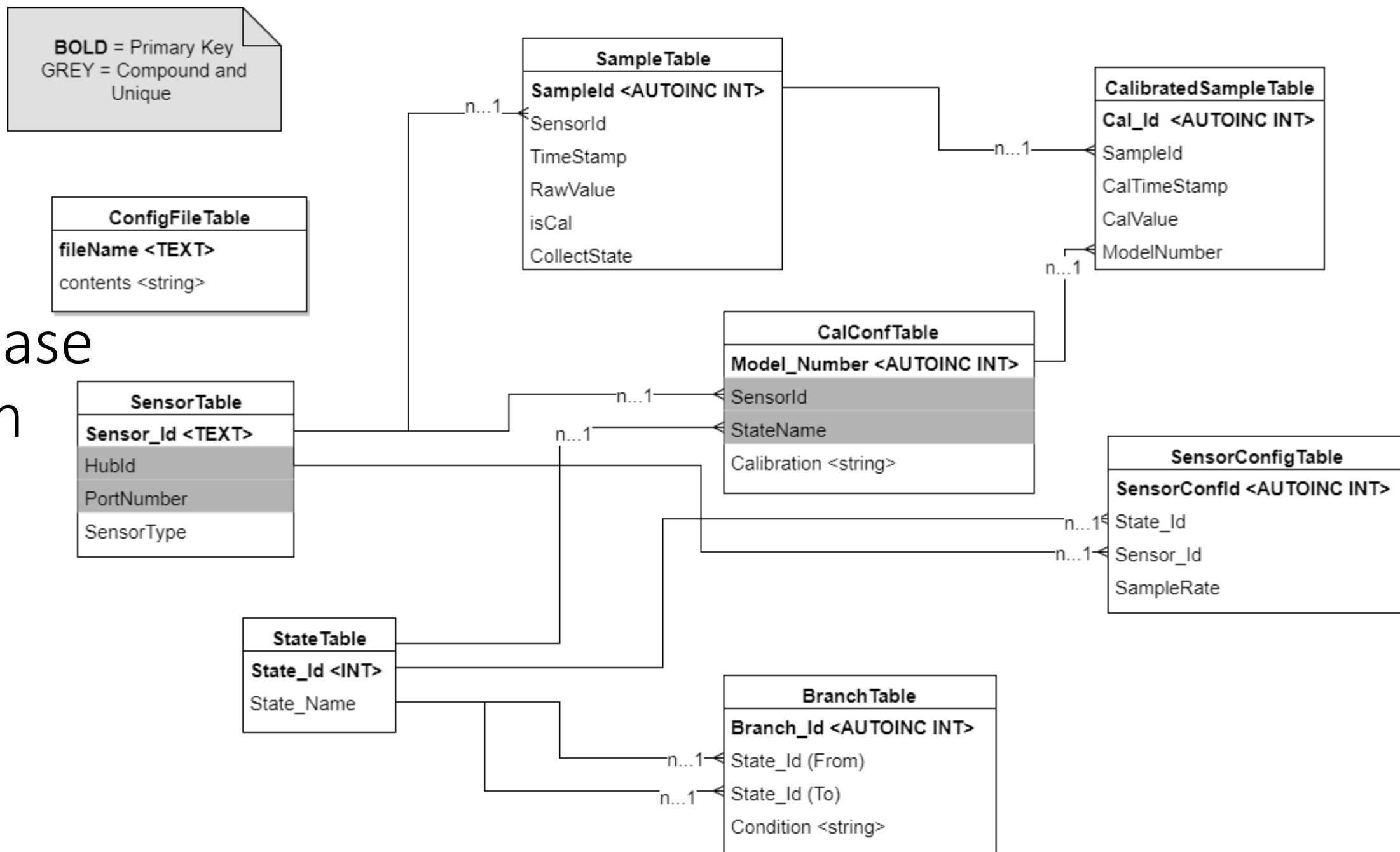
- Current:
 - Collection System
 - Data management System
 - Control System
 - Calibration System
 - Display/Export System
- Future:
 - Transforms
 - Data Graphing Tool
 - Indicators such as Alerts or emails from the SCADA system

Design

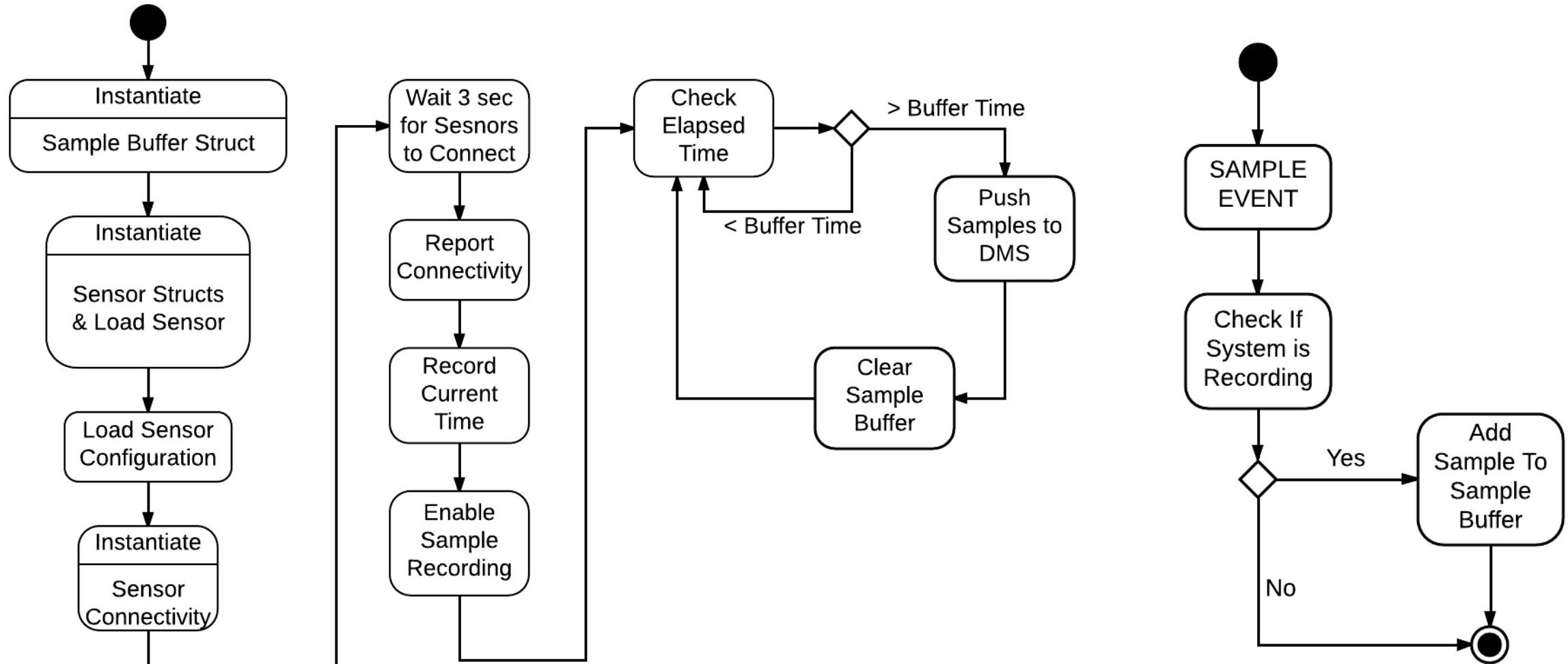


*DMS = Data Management System

Database Design

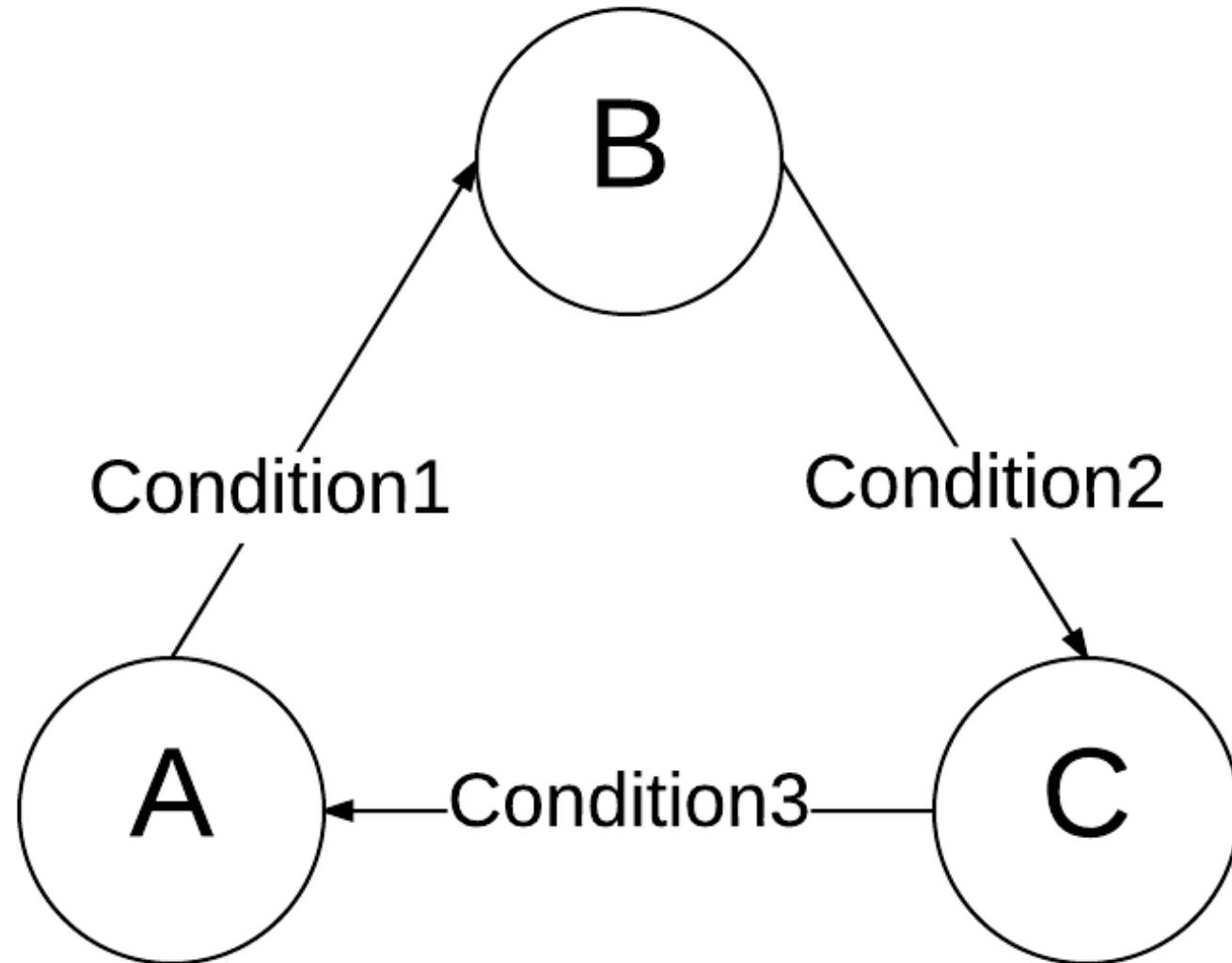


Collection System



Control System

- States
- Branches
- Conditions
 - Query



Calibration

- Selection of Sensors
- Selection of Data
- Calibration based on a state
 - Secondary Calibration implemented



Configurability

- Sensor Architecture
- State Architecture
- Collection
- Calibration



Configuration Format

```
#STATE syntax-> <STATE_1_Name>,<STATE_2_Name>,<STATE_3_Name> ...
STATES:
IDLE,RUN,STOP
#BRANCH syntax-> <STATE_FROM>,<STATE_TO>,<CONDITION>
#where <CONDITION> is <SENSOR_ID: SQL QUERY>
BRANCHES:
IDLE,RUN,s0:Select RawData from SampleTable where RawData > 2.0
RUN,STOP,s1:Select RawData from SampleTable where RawData > 2.5
STOP,IDLE,s1:Select RawData from SampleTable where RawData < 2.5
#SENSORS syntax-> <HUB_ID>,<PORT_NO>,<SENSOR_ID>
SENSORS:
4971941,1,s0
4953315,5,s1
#CALIBRATION syntax-> <SENSOR_ID>,<STATE_NAME>,<CALIBRATION>
#where <CALIBRATION> is <type:coeffs listed with ":" >
```

Configuration Format

CALIBRATION:

s0, IDLE, linear:2:1

s1, IDLE, linear:2:1

s0, RUN, linear:2:1

s1, RUN, linear:2:1

s0, STOP, quad:2:3:1

#COLLECTION syntax-> <SENSOR_ID>, <STATE_NAME>, <SAMPLING_PERIOD>

COLLECTION:

s0, IDLE, 2000

s1, IDLE, 2000

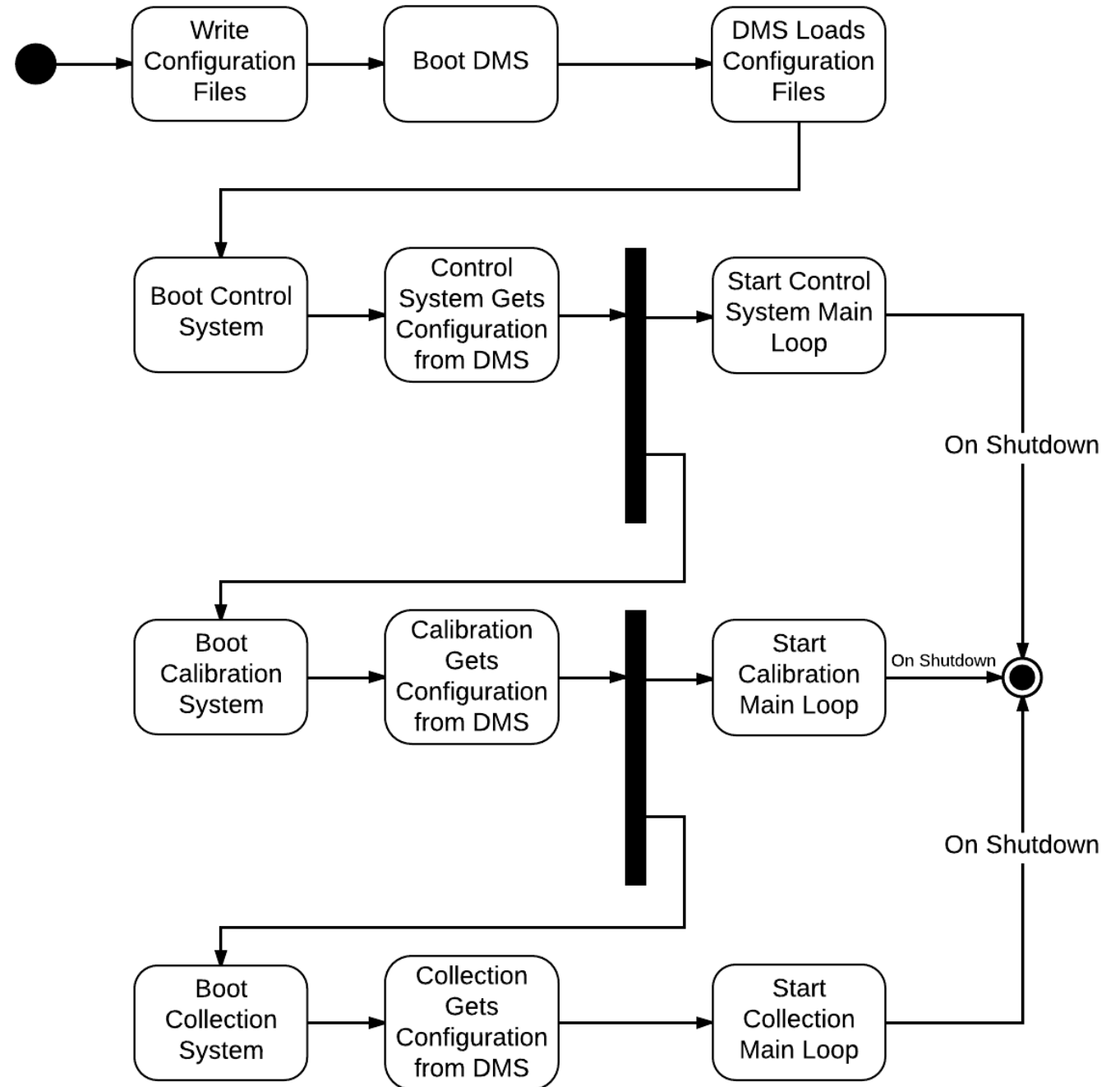
s0, RUN, 750

s1, RUN, 750

s0, STOP, 750

END

Boot Up Process

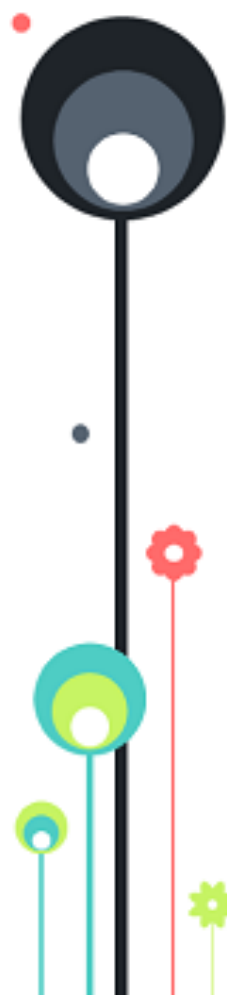




Talk is cheap.
Show me the code.

Linus Torvalds





Do you have any questions?

**THANK YOU FOR YOUR
ATTENTION !**